# University of Portsmouth

School of Creative Technologies

Final Year Project

# Evaluating the approachability of scientific papers towards creating a simulated particle physics scenario in C++ without prior background in the field

by

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## Abstract

This report investigates the accessibility of scientific literature from a software engineering perspective, using the creation of a sub-atomic particle physics simulation as a case study. Using physics research, a simulation will be made that aims to realistically reproduce the behaviour of the fundamental particles, including Protons, Neutrons, Electrons. Furthermore, the simulation will attempt to represent the four fundamental interactions: the strong, weak, electromagnetic, and gravitational forces.

Through the implementation of these concepts without prior academic or professional expertise, this project aims to evaluate the translatability and level of understanding needed to take the mathematics and concepts discussed into code.

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# Chapter 1

# Introduction

Software engineering has always been closely tied to physics and mathematics to the point where the former has become strongly associated with the latter. In some fields, such as game development which has a very large independant / solo community, this may be considered to be a barrier to entry for some. People may be dissuaded from learning how to program due to how heavy the mathematics might be, leading to the field missing out on great games made from the passion of talented people.

Resources such as research papers which are not explicitly designed with beginners-to-the-field, or game developers mystify the mathematics due to heavily verbose notation and content.

## 1.1 Project Aim

This project aims to evaluate and review how approachable research papers are for people who do not have a background in mathematics and physics, in creating a project.

The aim of this project is to create a plausible simulation of particle physics in C++, a project that will be heavily based in foundational physics.

## 1.2 Subjectivity Disclaimer

The evaluation of the accessibility and approachability of scientific papers and by extension the developer experience detailed in this report is inherently subjective, shaped by my own specific background in both particle physics (or lack thereof), and experience in programming, and interpretation of the resources used. As such, the challenges and successes documentated will vary for other developers with different backgrounds and levels of expertise in the two fields.

# Chapter 2

# Literature Review

## 2.1 Introduction

The purpose of a literature review is to identify existing areas of research relating to the given topic, alongside the any areas that are lacking in research literature. It also serves to provide an understanding of the topics that are intended to be discussed and expanded upon during the project.

## 2.2 Review of existing literature

Scientific literature regarding any subset of physics is traditionally written for those with a strong background of physics, mathematics, or research. Because of this, people outside of such disciplines, such as those in computer science including the game development fields, may be discouraged to use these resources due to the formatting, terminology or mathematic symbology that may unfamiliar to them. The result of this is developers who miss out on learning new techniques, algorithms, or optimalisation that may benefit their software or game. This literature review, and the following report evaluates foundational physics papers and their accessibility to those not discipined in physics or mathematics through the creation of a particle physics simulation in C++.

This review aims to source research within particle physics, and computer science.

### 2.2.1 Foundational Physics

This first category of literature consists of foundational physics papers, such as *A Standard Model Workbook* (Moore, 2024). While reports such as these serve to explain and give context to the bedrock of particle physics, they do so whilst presenting a developer experience barrier due to how its content is often dense mathematics.

From a software engineering perspective, the primary concern would be the use of abstract notation which, while logically might be familiar, are syntaxtically unorthodox. An example of this is $\sum$ which represents iterative accumulation, comparable that to a programmatical *for*

*loop*. Because of this notation, software and game developers would first need to translate the content from the paper into pseudo code.

### 2.2.2 Computational Physics

Computational literature such as *Computer Simulation in Physics and Engineering* (Steinhauser, 2012) are often based on the implementation of the theory established by foundational literature. Because of this, it is less mathematically dense, and more focal on the practical implementation of the functions and concepts introduced.

However, with this particular example, the argument could be made that it has the inverse problem than the foundational paper mentioned above, focusing heavily on the basics of computer science, such as explaining binary and arithmetic-in-code, rather than the actual implementation of the mathematics.

### 2.2.3 Developer Experience & Accessibility

The disparity between these two types of academic literature creates a gap between the the Developer Experience (**DX**), which can result in longer development time as considerable time may be required to 'reverse-engineer' the mathematics.

Furthermore, it should be noted that aims of the reseacher and developer may differ.

While the researcher may be showing the metholodgy used to achieve a specified result. The developer often times would also also aim for efficiency and scalablity in the algorithms used; something that the researcher would not have considered.

# Chapter 3

# Accessibility Metric System

To provide a basis of the subjective development experience, each source will be subjected to an analysis which tracks the frequency and severity of instances that the source is lacking in clarity and requires switching to other resources to understand, or add to, the context written.

### 3.0.1 Scoring Stystem

Each source will accumulate points depending on how far and how long I need to refer to other sources to understand the original.

Table 3.1: Complexity Severity Levels

| **Severity** | **Points** | **Criteria** | **Example** |
|---|---|---|---|
| Minor | 1 | Lexical issue. A word, phrase, or term that may not be immediately understood requiring brief additional research | 'Isospin', 'Alpha Decay' |
| Moderate | 2 | Notation barrier. Mathematical notation that is not explained requiring research into determining its meaning | $\sum$ |
| Major | 4 | Unexplained introduced concepts. Concepts that are mentioned but not elaborated which may require extensive further research | |
| Critical | 8 | Incomprehensive topic. A topic is incomprehensive and difficult to research or understand in the time given for research | A proof with no numerical constants, requiring learning how to calculate the values for that proof before even being able to use it. |

### 3.0.2 Evaluation Process

As the particle simulation is developed, each time the workflow is interrupted with needing to find context through additional resources, points will be accumulated and logged for each major resource, and a final **Approachability Score** will be created using the following equation:

$$A = \sum (P_{\text{minor}} + P_{\text{moderate}} + P_{\text{major}} + P_{\text{critical}}) \tag{3.1}$$

A lower Approachability Score indicates a more developer-friendly resource. Whereas a higher score suggests a more difficult resource for developers to use.

# Chapter 4

# Methodology

The methodology of this project will be double-faceted, focusing on the development of the simulation itself, and the concurrent audit of sources used to aide that development.

## 4.1 Simulation Architecture

The simulation is developed using C++ due to its high performance and stability, and memory management which will be important for handling large particle counts and their calculations efficiently.

- **Object-orientated design:** Each particle will be considered it's own object. Handling it's own position, radius, and systems such as rendering.
- **Environment:** Development is performed in a Linux environment using the GCC compiler.

The simulation also uses Dear ImGui (commonly referred to as *ImGui, imgui, or IMGUI*) to render a Graphical User Interface (GUI), this will provide the user statistics such as the number of particles, performance stats, alongside providing the user the ability to add or remove particles. (omar, 2026)

## 4.2 Approachability Audit

### 4.2.1 Data Collection Process

As additions are made to the simulation (e.g the gravitational force), the primary source that was used for that feature will be logged, and its approachability will be scored and logged based on the criteria shown in table 3.1.

### 4.2.2 Evaluation

At the end of the development of the simulation, the **Approachability Score *AS*** will be calculated and cross-referenced to the estimated time it had taken to implement the feature the source was used for.

## 4.3 Fundamental Forces

The first major system we will be implementing are the fundamental forces.

It should be noted that at this point, the project boilerplate has been written, such as the imgui interface, and the ability to place either a proton *(p)*, neutron *(n)*, electron *($e^-$)*, or photon *(ph)*, into the sandbox scene.

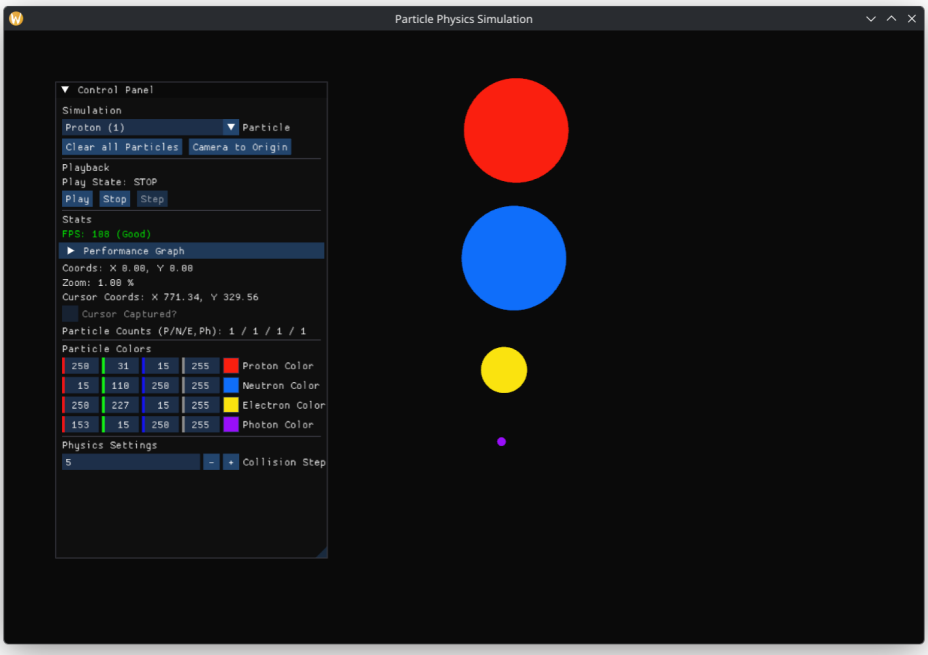


Figure 4.1: Initial project setup

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# References


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